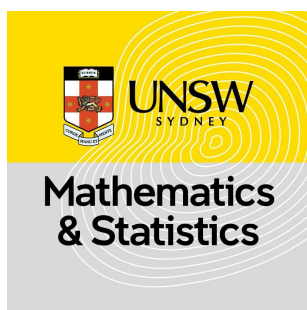




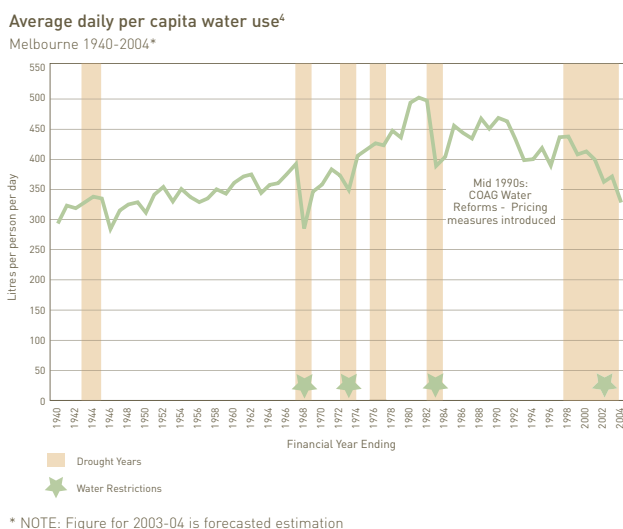
MATH3871/MATH5960 Bayesian Inference and Computation

Term 3, 2026

Tutorial Problems: Week 1

These exercises provide some revision in calculating expectations and other distributional results that you should know from other courses. The below results were (or will be) used in lectures.

1) Water consumption



In lectures, in the Melbourne average daily per-capita water use analysis, we modelled the discrete observations $x_1, \dots, x_n$ as independent draws from a $\text{Poisson}(\theta)$ distribution. Assuming a $\text{Gamma}(\alpha, \beta)$ prior distribution for θ , $\pi(\theta)$, which has a density function of

$$\pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} \exp(-\beta\theta), \quad \text{for } \alpha, \beta, \theta > 0,$$

we computed the posterior as a $\text{Gamma}(\alpha + \sum_{i=1}^n x_i, \beta + n)$ distribution. Recall that $\Gamma(a+1) = a\Gamma(a)$ for $a > 0$.

- By computing $\mathbb{E}[\theta]$, show that the posterior mean of θ is given by $\frac{\alpha + \sum_i x_i}{\beta + n}$.
- By computing $\text{Var}(\theta) = \mathbb{E}[\theta^2] - \mathbb{E}(\theta)^2$, show that the posterior variance of θ is given by $\frac{\alpha + \sum_i x_i}{(\beta + n)^2}$.
- Show that the resulting predictive distribution

$$\pi(y|x) = \int \pi(y|\theta)\pi(\theta|x)d\theta$$

for a future observation, y , is $\text{NegBin}\left(y \mid \alpha + \sum_i x_i, \frac{\beta+n}{\beta+n+1}\right)$, where the probability mass function of a Negative Binomial random variable with parameters $a > 0$, and $0 \leq p \leq 1$, is given by

$$\pi(y \mid a, p) = \binom{y+a-1}{y} (1-p)^y p^a.$$

2) Probability integral transformation (PIT)

We will make use of the PIT when developing inversion sampling. Suppose that a continuous random variable $X \sim f(x)$ has a density function $f_X(x)$ and distribution function $F_X(x)$. By computing its cumulative distribution function, show that the transformed random variable $U = F_X(X)$ follows a $U(0, 1)$ distribution.

3) Rock Strata (Bayes' Theorem)



Rock strata A and B are difficult to distinguish in the field. Through careful laboratory studies it has been determined that the only characteristic which might be useful in aiding discrimination is the presence or absence of a particular brachiopod fossil. In rock exposures of the size usually encountered, the probabilities of fossil presence are found to be as in the table below.

Stratum	Fossil present	Fossil absent
A	0.9	0.1
B	0.2	0.8

It is also known that rock type A occurs about four times as often as type B in this area of study (so that $\Pr(A) = 0.8$, $\Pr(B) = 0.2$).

- If one rock sample is taken, and the fossil found to be present, calculate the posterior distribution of rock types.
- If the geologist always classifies as A when the fossil is found to be present, and classifies as B when it is absent, what is the probability she will be correct in a future classification?

4) Coloured Balls



In the *Coloured Balls* example from lectures, there are 6 balls of unknown colour in a bag. Let C_i be the event that there are exactly i black balls in the bag, $i = 0, \dots, 6$. In lectures we specified our prior beliefs as to the number of black balls in each bag as: $\Pr(C_i) = \frac{1}{7}$ for each i .

- (a) Suppose that your prior knowledge is that the ball manufacturer makes 5 different ball colours, in equal proportions, and that 6 randomly chosen balls are put into each bag. That is, you believe that there is a $\frac{1}{5}$ probability that each ball is black. What are your prior probabilities for $C_0, \dots, C_6$?
- b) As per the question in the lecture: Three balls are drawn without replacement, and they are all observed to be black. Using the prior distribution in part a) what is the posterior probability of there being no black balls left in the bag?
- c) In what way does this change of prior (from $\Pr(C_i) = \frac{1}{7}$) affect the posterior probability of there being no black balls left in the bag?

5) Maximum likelihood as KL-divergence minimisation

This question is for those interested in an information-theoretic/optimisation-based interpretation of maximum likelihood estimation (and later, Bayesian inference: see later lectures).

Suppose that $x_1, \dots, x_n \sim p_{true}(x)$, $x_i \in \mathcal{X}$, from some unknown distribution $p_{true}(x)$, and we wish to fit a model $p(x|\theta)$; that is, estimate the parameter θ given the observed data. Define the Kullback-Leibler (KL) divergence between distributions $f(z)$ and $g(z)$ over some support $z \in \mathcal{Z}$ to be

$$\begin{aligned} D_{KL}(f||g) &= \int_{\mathcal{Z}} f(z) \log \left(\frac{f(z)}{g(z)} \right) dz \\ &= E_{f(z)} \left[\log \left(\frac{f(z)}{g(z)} \right) \right] = E_{f(z)}[\log f(z)] - E_{f(z)}[\log g(z)]. \end{aligned}$$

$D_{KL}(f||g) \geq 0$, taking larger values the more different f is to g , and only equals 0 when $f = g$.

- (a) Suppose that $\hat{\theta}$ is the maximum likelihood estimator of θ under the model $p(x|\theta)$. Show that

$$\hat{\theta} = \arg_{\theta \in \Theta} \min D_{KL}(p_{true}(x)||p(x|\theta)) \quad \text{as } n \rightarrow \infty.$$

That is, for large n , the maximum likelihood estimator is the value of θ that minimises the KL divergence between true and model data distributions.

(b) Conclude that for any $n < \infty$,

$$\hat{\theta} = \arg_{\theta \in \Theta} \min D_{KL}(\hat{p}_{true}^n \| p(x|\theta)),$$

where $\hat{p}_{true}^n(x|\theta) = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}(x)$ is the empirical estimate (measure) of $p_{true}(x)$, where $\delta_A(a) = 1$ if $a \in A$, and zero otherwise.